CMOS Analog Circuit Design

Analytical Design Procedure (P.E. Allen Methodology)

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1 Technology Parameters (1.2V CMOS Process)

The following BSIM4 extracted parameters are utilized for hand calculations. The channel length is fixed at $L = 0.5\ \mu$m for all transistors.

- $V_{DD} = 1.2$ V, $V_{SS} = 0$ V
- $K'_N = \mu_n C_{ox} = 457.2\ \mu$A/V^2, $\quad K'_P = \mu_p C_{ox} = 63.8\ \mu$A/V^2
- $V_{TN0} = 0.343$ V, $\quad |V_{TP0}| = 0.235$ V
- $\lambda_N = 0.0528$ V^{-1}, $\quad \lambda_P = 0.0469$ V^{-1}

2 Design 1: 5-T Differential OTA (Current Mirror Load)

2.1 Specifications

- Slew Rate (SR) ≥ 100 V/μs with Load Capacitance $C_L = 1$ pF
- Open-Loop Voltage Gain (A_v) ≥ 60 V/V
- Input Common-Mode Range (ICMR): $0.85\text{ V} \leq V_{IC} \leq 1.05$ V

2.2 Design Procedure

Step 1: Calculate Tail Current (I_{SS})
From Slew Rate requirements:

$$I_{SS} = SR \times C_L = 100 \times 10^6 \times 10^{-12} = 100\ \mu\text{A} \tag{1}$$

Assuming perfect matching, the current in the input branches is $I_{D1} = I_{D2} = I_{D3} = I_{D4} = 50\ \mu$A. **Step 2: Check Output Resistance (R_{out})**

$$R_{out} = \frac{1}{(\lambda_N + \lambda_P) I_{D1}} = \frac{1}{(0.0528 + 0.0469) \times 50\mu} = 200.6\ \text{k}\Omega \tag{2}$$

Step 3: Design Input Pair (M1, M2) for Target Gain

$$g_{m1} = \frac{A_v}{R_{out}} = \frac{60}{200.6\text{k}} = 299.1\ \mu\text{S} \tag{3}$$

Using the transconductance equation $g_{m1} = \sqrt{2K'_N(W/L)_1 I_{D1}}$:

$$(W/L)_1 = \frac{g_{m1}^2}{2K'_N I_{D1}} = \frac{(299.1\mu)^2}{2 \times 457.2\mu \times 50\mu} = 1.96 \tag{4}$$

$\Rightarrow W_1 = W_2 = 1.96 \times 0.5 = \mathbf{0.98}\ \mu$m. **Step 4: Design Load Pair (M3, M4) for Upper ICMR**

$$V_{IC(max)} = V_{DD} - V_{SG3} + V_{TN0} \implies 1.05 = 1.2 - V_{SG3} + 0.343 \implies V_{SG3} = 0.493 \text{ V} \tag{5}$$

$$V_{SG3} = \sqrt{\frac{2I_{D3}}{K'_P(W/L)_3}} + |V_{TP0}| \implies 0.493 = \sqrt{\frac{100\mu}{63.8\mu(W/L)_3}} + 0.235 \tag{6}$$

Solving for $(W/L)_3$ gives $(W/L)_3 = 23.54$.
$\Rightarrow W_3 = W_4 = 23.54 \times 0.5 = \mathbf{11.77}\ \mu$m. **Step 5: Design Tail Current Source (M5) for Lower ICMR**

$$V_{GS1} = \sqrt{\frac{2I_{D1}}{K'_N(W/L)_1}} + V_{TN0} = \sqrt{\frac{100\mu}{457.2\mu \times 1.96}} + 0.343 = 0.677 \text{ V} \tag{7}$$

$$V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1} \implies 0.85 = 0 + V_{DS5(sat)} + 0.677 \implies V_{DS5(sat)} = 0.173 \text{ V} \tag{8}$$

$$(W/L)_5 = \frac{2I_{SS}}{K'_N(V_{DS5(sat)})^2} = \frac{2 \times 100\mu}{457.2\mu \times (0.173)^2} = 14.6 \tag{9}$$

$\Rightarrow W_5 = 14.6 \times 0.5 = \mathbf{7.3}\ \mu$m. Bias Voltage $V_{bias} = V_{TN0} + V_{DS5(sat)} = 0.343 + 0.173 = \mathbf{0.516}$ V. **Step 6: Calculate Expected Gain-Bandwidth Product (GBW)**
The unity-gain frequency (GBW) is determined by the input transconductance and the load capacitance:

$$GBW = \frac{g_{m1}}{2\pi C_L} = \frac{299.1\ \mu\text{S}}{2\pi \times 1\ \text{pF}} = 47.6 \text{ MHz} \tag{10}$$

2.3 Comparison and Discussion

A comparison between a heuristic engineering approach (arbitrarily selecting V_{OV}) and the deterministic P.E. Allen approach (forcing dimensions based on strict ICMR specifications) is shown below: **Discussion:**

Component (schematic name)	Heuristic Design	P.E. Allen Method
M0, M1 (Input)	$W = 1.08\ \mu$m	$\mathbf{W = 0.98}\ \mu$m
M2, M3 (Load)	$W = 12.55\ \mu$m	$\mathbf{W = 11.77}\ \mu$m
M4 (Tail)	$W = 5.45\ \mu$m	$\mathbf{W = 7.3}\ \mu$m

Table 1: Comparison of Transistor Widths ($L = 0.5\ \mu$m)

The deterministic procedure developed by P.E. Allen is superior because it mathematically guarantees that the amplifier will **never experience signal clipping** as long as the input common-mode voltage remains within the specified range (0.85 V $\leq V_{IC} \leq$ 1.05 V). To achieve this, the tail current source (M4 in the schematic, M5 in the textbook) is deliberately sized larger (7.3 μm vs 5.45 μm) to minimize its saturation voltage $V_{DS(sat)}$ (or V_{OV}). This preserves voltage headroom for the differential pair operating above it.

3 Design 2: Two-Stage CMOS Operational Amplifier

3.1 Specifications

- Load Capacitance $C_L = 2$ pF
- Slew Rate (SR) ≥ 50 V/μs
- Gain-Bandwidth Product (GBW) ≥ 50 MHz
- Phase Margin (PM) $\approx 60^\circ$
- ICMR: 0.85 V $\leq V_{IC} \leq$ 1.05 V

3.2 Design Procedure

Step 1: Choose Compensation Capacitor (C_c)
For $PM \geq 60^\circ$, select $C_c \geq 0.22C_L$. Thus, we choose $\mathbf{C_c = 1}$ pF. **Step 2: Calculate Tail Current (I_5)**

$$I_5 = SR \times C_c = 50 \times 10^6 \times 10^{-12} = 50\ \mu\text{A} \tag{11}$$

To ensure robust performance, we choose $I_5 = \mathbf{100}\ \mu$A. Therefore, $I_{D1} = I_{D2} = 50\ \mu$A. **Step 3: Design Input Pair (M1, M2) for GBW**

$$GBW = \frac{g_{m1}}{2\pi C_c} \implies g_{m1} = 50 \times 10^6 \times 2\pi \times 10^{-12} = 314.16\ \mu\text{S} \tag{12}$$

$$(W/L)_1 = \frac{g_{m1}^2}{2K_N' I_{D1}} = \frac{(314.16\mu)^2}{2 \times 457.2\mu \times 50\mu} = 2.158 \tag{13}$$

$\Rightarrow W_1 = W_2 = 2.158 \times 0.5 = \mathbf{1.08}\ \mu$m. **Step 4: Design Load Pair (M3, M4) for Upper ICMR**
Same requirement as Design 1 ($V_{IC(max)} = 1.05$ V).
$\Rightarrow W_3 = W_4 = \mathbf{11.77}\ \mu$m. **Step 5: Design Tail Current Source (M5) for Lower ICMR**

$$V_{GS1} = \sqrt{\frac{2I_{D1}}{K_N'(W/L)_1}} + V_{TN0} = \sqrt{\frac{100\mu}{457.2\mu \times 2.158}} + 0.343 = 0.661\ \text{V} \tag{14}$$

$$V_{IC(min)} = V_{SS} + V_{DS5(sat)} + V_{GS1} \implies 0.85 = 0 + V_{DS5(sat)} + 0.661 \implies V_{DS5(sat)} = 0.189\ \text{V} \tag{15}$$

$$(W/L)_5 = \frac{2 \times 100\mu}{457.2\mu \times (0.189)^2} = 12.24 \tag{16}$$

$\Rightarrow W_5 = 12.24 \times 0.5 = \mathbf{6.12}\ \mu$m. Bias Voltage $V_{bias} = V_{TN0} + V_{DS5(sat)} = 0.343 + 0.189 = \mathbf{0.532}$ V. **Step 6: Design Second Stage (M6) for Phase Margin and Zero Systematic Offset**
To prevent systematic offset, M6 and M4 must have equal current densities ($V_{SG6} = V_{SG4}$):

$$\frac{I_{D6}}{(W/L)_6} = \frac{I_{D4}}{(W/L)_4} \implies (W/L)_6 = (W/L)_4 \frac{I_{D6}}{I_{D4}} \tag{17}$$

For $PM \geq 60^\circ$, place the non-dominant pole $p_2 \geq 2.2 \times GBW$.

$$\frac{g_{m6}}{C_L} \geq 2.2(2\pi\ GBW) \implies g_{m6} \geq 2.2 \times 314.16\mu \times \frac{2\text{pF}}{1\text{pF}} = 1382\ \mu\text{S} \tag{18}$$

Combining both constraints, $I_{D6} \geq 178\ \mu$A. We choose $I_{D6} = \mathbf{200}\ \mu$A.

$$(W/L)_6 = 23.54 \times \frac{200\mu}{50\mu} = 94.16 \implies W_6 = 94.16 \times 0.5 = \mathbf{47.08}\ \mu\text{m} \tag{19}$$

Step 7: Design Second Stage Load (M7)
M7 is mirrored from M5. It must sink $I_{D7} = I_{D6} = 200\ \mu$A.

$$(W/L)_7 = (W/L)_5 \frac{I_{D7}}{I_{D5}} = 12.24 \times \frac{200\mu}{100\mu} = 24.48 \implies W_7 = 24.48 \times 0.5 = \mathbf{12.24}\ \mu\text{m} \tag{20}$$

Step 8: Calculate Expected DC Gain (A_v)
The total DC gain is the product of the first-stage and second-stage gains ($A_v = A_{v1} \times A_{v2}$).

$$R_{out1} = \frac{1}{(\lambda_N + \lambda_P) I_{D1}} = \frac{1}{0.0997 \times 50\mu} = 200.6\,\text{k}\Omega \implies A_{v1} = g_{m1} R_{out1} = 314.16\mu \times 200.6\text{k} = 63.0\ \text{V/V} \tag{21}$$

For the second stage, g_{m6} is derived from I_{D6} and $(W/L)_6$:

$$g_{m6} = \sqrt{2K'_P (W/L)_6 I_{D6}} = \sqrt{2 \times 63.8\mu \times 94.16 \times 200\mu} = 1550\ \mu\text{S} \tag{22}$$

$$R_{out2} = \frac{1}{(\lambda_N + \lambda_P) I_{D6}} = \frac{1}{0.0997 \times 200\mu} = 50.15\,\text{k}\Omega \implies A_{v2} = g_{m6} R_{out2} = 1550\mu \times 50.15\text{k} = 77.7\ \text{V/V} \tag{23}$$

$$A_{v(total)} = A_{v1} \times A_{v2} = 63.0 \times 77.7 = 4895\ \text{V/V} \implies 20\log_{10}(4895) = \mathbf{73.8}\ \text{dB} \tag{24}$$

3.3 Auxiliary Parameters

1. Power Dissipation (P_{diss}):
The total current consumed by the two-stage operational amplifier is the sum of the bias currents in both stages.

$$I_{total} = I_{tail1} + I_{tail2} = 100\ \mu\text{A} + 200\ \mu\text{A} = 300\ \mu\text{A} \tag{25}$$

$$P_{diss} = I_{total} \times (V_{DD} - V_{SS}) = 300\ \mu\text{A} \times 1.2\ \text{V} = \mathbf{360}\ \mu\text{W} \tag{26}$$

2. Output Voltage Swing (OVS):
The maximum and minimum output voltages are limited by the saturation voltages of the output stage transistors (M6 and M7).

$$|V_{DS6(sat)}| = \sqrt{\frac{2I_{D6}}{K'_P (W/L)_6}} = \sqrt{\frac{2 \times 200\mu}{63.8\mu \times 94.16}} = 0.258\ \text{V} \tag{27}$$

$$V_{DS7(sat)} = \sqrt{\frac{2I_{D7}}{K'_N (W/L)_7}} = \sqrt{\frac{2 \times 200\mu}{457.2\mu \times 24.48}} = 0.189\ \text{V} \tag{28}$$

Therefore, the output voltage can swing from $V_{min} = V_{SS} + V_{DS7(sat)} = \mathbf{0.189}$ V to $V_{max} = V_{DD} - |V_{DS6(sat)}| = 1.2 - 0.258 = \mathbf{0.942}$ V.

3. Systematic Input Offset Voltage ($V_{OS(sys)}$):
In Step 6, the dimensions of M6 were explicitly calculated to ensure equal current densities between M6 and M4: $(W/L)_6/(W/L)_4 = I_{D6}/I_{D4} = 4$. By satisfying this constraint ($V_{SG6} = V_{SG4}$), the DC output voltage of the first stage perfectly matches the required bias for the second stage. This mathematical balance yields a theoretical Systematic Input Offset Voltage of **0** V.

4. Right-Half Plane (RHP) Zero Check:
The compensation capacitor C_c introduces a feedforward path that creates an RHP zero, degrading the phase margin.

$$z_1 = \frac{g_{m6}}{2\pi C_c} = \frac{1550\ \mu\text{S}}{2\pi \times 1\ \text{pF}} = 246.7\ \text{MHz} \tag{29}$$

Since $z_1 \approx 4.9 \times GBW$, it will introduce some phase shift at the unity-gain frequency ($\approx -11.5^\circ$). To completely eliminate this effect and improve Phase Margin, a nulling resistor (R_z) can be added in series with C_c:

$$R_z = \frac{1}{g_{m6}}\left(1 + \frac{C_L}{C_c}\right) = \frac{1}{1550\mu}\left(1 + \frac{2\text{pF}}{1\text{pF}}\right) = 1.93\ \text{k}\Omega \tag{30}$$

5. Mirror Pole Check (p_3):
The current mirror load (M3, M4) introduces a parasitic pole at the drain of M1/M3.

$$g_{m3} = \sqrt{2K_P'(W/L)_3 I_{D3}} = \sqrt{2 \times 63.8\mu \times 23.54 \times 50\mu} = 122.5\ \mu\text{S} \tag{31}$$

Assuming parasitic capacitance $C_m \approx 0.2$ pF at this node, the mirror pole is located at:

$$p_3 = \frac{g_{m3}}{2\pi C_m} = \frac{122.5\ \mu\text{S}}{2\pi \times 0.2\ \text{pF}} \approx 97.5\ \text{MHz} \tag{32}$$

Since $p_3 < 2 \times GBW$, this pole will also slightly degrade the phase margin. This explains why the simulated Phase Margin was 50.3° instead of the ideal 60°.